AI 101 for Mechanical Engineers · Module 4

Self-supervised targets come from the data

Lesson 4.2

What you will learn

> Explain how self-supervised learning constructs targets from observed data, follow a masked-signal reconstruction task, and distinguish model architecture from learning feedback.

Lesson overview

Module 4 · Lesson 2

Self-supervised targets come from the data

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1 Hide part of a signal, then reconstruct it

Training targets are created from the observed data using a rule.

1 Observation
The original measured signal.



2 Constructed input
The signal with a section hidden.



3 Constructed target
The original values in the hidden section.



4 Objective
Reduce the disagreement between the estimate and the target.



Masked input → Model → Estimated hidden values → Compare with retained hidden values

Keep the hidden target separate from the model input. Another example: predict the next token from preceding tokens.

2 Structure and feedback are different

Architecture
How the model is structured.

Learning setup
How training feedback is supplied.

A neural network can use:

Supervised learning	supplied targets
Self-supervised learning	targets created from data
Reinforcement learning	reward feedback

Knowing the architecture does not tell you the learning setup.

Key takeaway

Self-supervised learning still has targets, an objective and a loss.
Reconstructing a signal does not provide a confirmed fault label.

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Self-supervised targets come from the data

- Targets come from the data via a target-generation rule
- BERT hides some words and predicts them from surrounding words [29]
- Autoregressive language models predict the next token [67]
- Signal example: teaching illustration, not a result in those papers

No manual labels does not mean no objective

Still present

- Inputs
- Targets
- A loss measuring disagreement

What differs

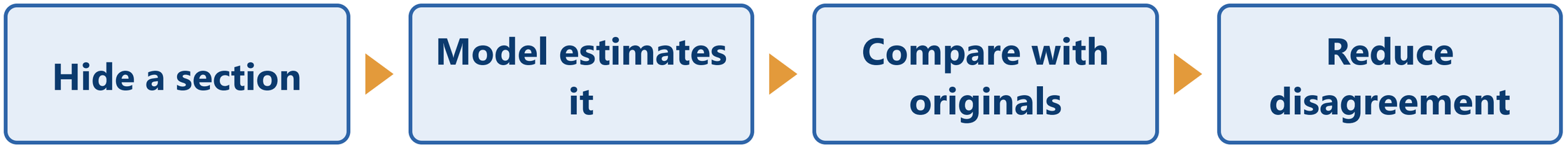
- How the targets are obtained
- Not whether targets exist

Masked-signal reconstruction has four parts

Part	Content
Observation	Original measured signal
Constructed input	Signal with a section deliberately hidden
Constructed target	Original values in the hidden section
Objective	Reduce reconstructed-versus-original disagreement

> Manual annotation of each hidden value is not required.

The original signal supplies the training reference



- Learned representations may later be adapted to labelled tasks
- The reconstruction target is not a manually confirmed fault label

Architecture and learning setup are separate

Architecture

- Structure of a model
- e.g. a neural network [4]

Learning setup

- How training feedback is supplied
- Manual labels, self-supervised targets or reinforcement feedback

> Naming a neural network does not identify its learning setup. State task, feedback and model separately.

CHECK YOUR UNDERSTANDING

In masked-signal reconstruction, where does the target come from?

Answer before moving on.

CHECK YOUR UNDERSTANDING

In masked-signal reconstruction, where does the target come from?

> From the original measured values in the section deliberately hidden from the model’s input.

CHECK YOUR UNDERSTANDING

Does a dataset without manually supplied fault labels imply that training has no objective?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does a dataset without manually supplied fault labels imply that training has no objective?

> No. A self-supervised task constructs its own targets from observed data and evaluates a loss against those targets.

Key takeaways

- Self-supervised targets are constructed from observed data
- Masking changes the input; original values remain the targets
- Training still has an objective and feedback
- Model architecture does not determine a unique learning setup

References

[4] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.

[29] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in Proc. 2019 Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol. (NAACL-HLT), 2019, pp. 4171–4186. doi: 10.18653/v1/N19-1423

[67] T. B. Brown et al., “Language models are few-shot learners,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020.

Numbers match the course reference list. The written lesson is the authoritative text.